Department of Computer Science & Engineering

Software Engineering

UE24CS341A

Lab 7: Fuzz Testing Submission

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Section 1

Section 1 - Objective and setup

Task: Use Hypothesis to discover input validation bugs in the assigned parking application, fix them, and verify the fixes.

Assigned repository: https://github.com/SETAPESU26/L6_L7_19_parking_fee (Lab 7/). Source commit: fef20d42191cdd992b14c5717db612b929c2cfd0.

The repository filenames parking_fee.py and test_parking_fee.py correspond to processor.py and test_processor.py in the handout.

Environment used: 3.14.6 (main, Aug 10 2026, 06:36:12) [GCC 16.1.1 20260728]. Dependencies: pytest 9.2.0 and Hypothesis 6.170.15.

Commands: python -m pip install -r requirements.txt; python parking_fee.py; python -m pytest test_parking_fee.py.

Method: Record the original application output, complete the three starter properties, retain their failures, fix each helper, then rerun both those properties and the expanded suite.

The final suite has 79 helper test instances (14 Hypothesis property instances and 65 explicit regression cases), plus nine CLI cases: 88 passing tests in total. A property instance may execute hundreds of generated examples; the pytest test count is not the generated-input count.

Arbitrary Unicode is useful for finding crashes, but rarely resembles a valid time range or plate. Structured strategies also generate valid values and check their exact outputs against independent expectations. Hypothesis shrinking reduces a failure to a small example for diagnosis.

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Section 2

Section 2 - Step 2: Run the original application

Input: 09:30-11:45, car, ka01ab1234. The original program prints 135 minutes, rate 30, and KA 01 AB 1234.

An empty time input causes ValueError and stops the program before the remaining prompts. The baseline application is preserved unchanged in baseline/parking_fee.py.

Screenshot 1: Original application, valid input

```
d3va@legion:baseline$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline
d3va@legion:baseline$ python -u parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): car
Hourly Rate: 30

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): ka01ab1234
Formatted Plate: KA 01 AB 1234
d3va@legion:baseline$ printf 'Exit code: %s\n' "$?"
Exit code: 0
d3va@legion:baseline$
```

Screenshot 2: Original application, invalid input

```
d3va@legion:baseline$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline
d3va@legion:baseline$ python -u parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45):
Traceback (most recent call last):
  File "/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline/parking_fee.
py", line 33, in <module>
    main()
    ~~~~^^
  File "/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline/parking_fee.
py", line 19, in main
    result = minutes_parked(user_input)
  File "/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline/parking_fee.
py", line 4, in minutes_parked
    entry, exit_time = times.split('-')
    ^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
d3va@legion:baseline$ printf 'Exit code: %s\n' "$?"
Exit code: 1
d3va@legion:baseline$
```

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Section 3

Section 3 - Steps 3 and 4: Fuzz tests expose failures

Each starter property uses @given(st.text() | st.none()) with max_examples=500 and deterministic generation. It checks that the function does not throw and that any returned value has a valid basic shape.

All three tests fail against the original code. Hypothesis reports seven distinct failing outcomes across those three tests: time None and empty text; rate None and empty text; plate None, empty text, and '00'.

The full tracebacks and minimized inputs are preserved in reports/before/pytest.txt. The screenshot shows a separate run with traceback display disabled for readability; no failures are hidden from the test result.

Screenshot 3: Three failing Hypothesis properties

```
d3va@legion:baseline$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7/baseline
d3va@legion:baseline$ python -m pytest --import-mode=importlib fuzz_snapshot.py -q --tb=no --co
lor=yes
FFF                                                                      [100%]
=================================== short test summary info ====================================
FAILED fuzz_snapshot.py::test_minutes_parked_no_crash - ExceptionGroup: Hypothesis found 2 dist
inct failures. (2 sub-exceptions)
FAILED fuzz_snapshot.py::test_hourly_rate_no_crash - ExceptionGroup: Hypothesis found 2 distinc
t failures. (2 sub-exceptions)
FAILED fuzz_snapshot.py::test_format_plate_no_crash - ExceptionGroup: Hypothesis found 3 distin
ct failures. (3 sub-exceptions)
3 failed in 0.42s
d3va@legion:baseline$ printf 'Exit code: %s\n' "$?"
Exit code: 1
d3va@legion:baseline$
```

Section 4

Section 4 - Bug analysis and input policy

minutes_parked: None raises AttributeError; empty text raises ValueError. Directed checks also show that 25:00-26:00 returns 60, 09:60-10:00 returns 0, and 11:00-10:00 returns -60.

hourly_rate: None raises AttributeError and unknown or empty text raises KeyError.

format_plate: None raises AttributeError, empty text raises IndexError, and '00' produces two digits followed by three spaces. A long plate such as KA01AB123456 is silently truncated.

Contract: Invalid inputs return None. Time fields must use ASCII HH:MM-HH:MM, valid clock ranges, and same-day forward order. Equal times return 0; overnight stays need a date and are rejected here.

Rates retain bike=10, car=30, truck=60, with surrounding whitespace and case normalization. Plates accept the lab's AA99AA9999 syntax in compact or single-space grouped form and produce uppercase AA 99 AA 9999.

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Section 5

Section 5 - Step 5: Fix and rerun the application

The helpers validate types, complete formats, clock ranges, and same-day order. The CLI explains invalid entries and continues; EOF and Ctrl+C exit cleanly. A valid zero is preserved, and the original sample outputs remain correct.

Screenshot 4: Fixed application, valid input

```
d3va@legion:Lab-7$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7
d3va@legion:Lab-7$ python -u parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45): 09:30-11:45
Minutes Parked: 135

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): car
Hourly Rate: 30

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): ka01ab1234
Formatted Plate: KA 01 AB 1234
d3va@legion:Lab-7$ printf 'Exit code: %s\n' "$?"
Exit code: 0
d3va@legion:Lab-7$
```

Screenshot 5: Fixed application, invalid input

```
d3va@legion:Lab-7$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7
d3va@legion:Lab-7$ python -u parking_fee.py
==== Running minutes_parked ====
Enter entry and exit time (e.g. 09:30-11:45):
Invalid time range. Use HH:MM-HH:MM in same-day order.

==== Running hourly_rate ====
Enter vehicle type (bike/car/truck): hovercraft
Invalid vehicle type. Choose bike, car, or truck.

==== Running format_plate ====
Enter a number plate (e.g. ka01ab1234): x
Invalid number plate. Use KA01AB1234 or KA 01 AB 1234 format.
d3va@legion:Lab-7$ printf 'Exit code: %s\n' "$?"
Exit code: 0
d3va@legion:Lab-7$
```

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Section 6

Section 6 - Step 6: Rerun the tests

The same three starter properties now pass. The expanded suite also checks exact valid results, time boundaries and ordering, case/whitespace normalization, plate idempotence, malformed formats, unsupported Python types, and command-line behavior.

Generated valid examples use independent expected values, including datetime arithmetic for durations and explicit known rate values. Returning None for every input would fail these properties.

Final pytest result: 88 passed in 4.61s. Raw results are saved in reports/after/. The screenshot below shows the complete test suite, including CLI regression tests.

Screenshot 6: Final complete test suite

```
d3va@legion:Lab-7$ pwd
/home/d3va/Downloads/Academics_General/Sem-5/SE/lab7/SELABS/Lab-7
d3va@legion:Lab-7$ python -m pytest -q --color=yes
.......................................................................................... [ 98%]
.                                                                                         [100%]
88 passed in 4.61s
d3va@legion:Lab-7$ printf 'Exit code: %s\n' "$?"
Exit code: 0
d3va@legion:Lab-7$
```

Section 7

Section 7 - Deliverables and evidence

PDF: PES1UG24CS145_SE_Lab7.pdf. Fixed application: parking_fee.py. Completed Hypothesis suite: test_parking_fee.py. Additional CLI tests: test_cli.py.

Supporting files: INPUT_CONTRACT.md, BUG_ANALYSIS.md, unchanged baseline, raw reports, screenshot PNGs, and tools to reproduce the evidence and this PDF.

Screenshots are direct Grim captures of a live fullscreen Foot terminal using the current user theme. Commands, typed inputs, working directories, results, and exit codes are visible in the terminal. Capture manifests record UTC timestamps, window geometry, PNG hashes, and matching terminal transcript hashes.

Fuzzing identified crashes and malformed output with small examples. Boundary checks added silent validation failures. The revised suite checks both rejection and correct handling of valid inputs; passing it does not prove correctness for all possible inputs.